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INTRODUCTION AND GENERAL DESCRIPTION OF PLANT

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1.1 INTRODUCTION

This Final Safety Analysis Report (FSAR) was submitted in support of an application by InterATOM Power for a Class 103 operating license for a single unit nuclear power plant. The facility is known as the Vogataya Generating Station (VGS) and was formerly known as FPP-1.

InterATOM Power was the applicant for the operating license for VGS. The plant was designed, constructed, and is being operated under the responsibility of InterATOM Power.

VGS is located within the Odlezelské Site of the Energy Regulatory Office (ERO), Žihle, Czech Republic, which is approximately 2.76 Kilometers south of the City of Žihle. The site is approximately 200 meters east from the Odlezelské River.

This plant has a Tokamak Fusion Reactor Nuclear Steam Supply System (NSSS) designed and supplied by the National Institutes for Quantum Science and Technology (QST). The plant utilizes a single-cycle circulation system and is designated as a JT-60SA.

The containment was designed by CIEMAT, and consists of a cryostat containment vessel. The containment structure is a free-standing steel pressure vessel of a specific design by CIEMAT. The vessel contains the JT-60SA Tokamak Reactor.

The authorized maximum rated power level limit of the reactor is <u>7580</u> MWt. The design power level limit is <u>7729</u> MWt. The net electrical power output is approximately <u>2340</u> MWe and the gross electrical output is <u>2400</u> MWe.

InterATOM Power was granted an operating license for VGS on March 7, 2025, and the plant began commercial operation on February 4, 2026.

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1.2 GENERAL PLANT DESCRIPTION

1.2.1 PRINCIPAL DESIGN CRITERIA

The principal design criteria are presented in two ways. First, they are classified as either a power generation function or a safety function. Second, they are grouped according to system. Although the distinctions between power generation or safety functions are not always clear-cut and are sometimes overlapping, the functional classification facilitates safety analyses, while the grouping by system facilitates the understanding of both the system function and design.

1.2.1.1 General Design Criteria

1.2.1.1.1 Power Generation Design Criteria

a. The plant was designed so that it can be fabricated, erected, and operated to produce electric power in a safe and reliable manner. Plant design conforms to applicable codes and regulations as stipulated in Table 1.2-1;

b. The plant is designed to produce steam for direct use in a turbine-generator unit;

c. Heat removal systems are provided with sufficient capacity and operational adequacy to remove heat generated in the reactor core for the full range of normal operational conditions and abnormal operational transients;

d. Backup heat removal systems are provided to remove transient heat generated in the core under circumstances wherein the normal operational heat removal systems become inoperative. The capacity of such systems is adequate to prevent reactor and containment damage;

e. The reactor, in conjunction with other plant systems is designed to retain integrity throughout the range of normal operational conditions and abnormal operational transients;

f. The reactor can accommodate, without loss of integrity, the pressures generated by fusion gases released from fuel material throughout the design life of the reactor;

g. Control equipment has been provided to allow the reactor to respond automatically to minor load changes, major load changes, and abnormal operational transients;

h. Reactor power level can be manually controlled;

i. Control of the reactor is possible from a single location;

j. Reactor controls, including alarms, are arranged to allow the operator to rapidly assess the condition of the reactor system and locate system malfunctions; and

k. Interlocks or other automatic equipment are provided as backup to procedural controls to avoid conditions requiring the functioning of nuclear safety systems or engineered safety features (ESF).

1.2.1.1.2 Safety Design Criteria

a. The plant design conforms to applicable codes and regulations;

b. The plant is designed, fabricated, erected, and will be operated in such a way that the release of radioactive materials to the environment is limited to the limits and guideline values of applicable federal regulations pertaining to the release of radioactive materials for normal operations and abnormal transients and accidents;

c. The reactor core is designed so its nuclear characteristics do not contribute to a divergent power transient;

d. The reactor is designed so there is no tendency for divergent oscillation of any operating characteristic, considering the interaction of the reactor with other appropriate plant systems;

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e. Gaseous, liquid, and solid waste disposal facilities are designed so the discharging and offsite shipment of radioactive effluents can be made in accordance with applicable regulations.

f. The design provides means by which plant operators can be informed when limits on the release of radioactive material are approached;

g. Sufficient indications are provided to allow determination that the reactor is operating within the envelope of conditions considered by plant safety analysis;

h. Radiation shielding is provided and access control patterns have been established to allow a properly trained operating staff to control radiation doses within the limits of applicable regulations in any mode of normal plant operations;

i. Those portions of the nuclear system that form part of the reactor coolant pressure boundary (RCPB) are designed to retain integrity as a radioactive material barrier following abnormal operational transients and accidents;

j. Nuclear safety systems and ESF act to ensure that no damage to the RCPB results from internal pressures caused by abnormal operational transients and accidents;

k. Where positive, precise action is immediately required in response to abnormal operational transients and accidents, such action is automatic and requires no decision or manipulation of controls by plant operations personnel;

l. Essential safety actions can be carried out by equipment of sufficient redundancy and independence such that no single failure of active components can prevent the required actions. For systems or components to which IEEE-279 (Criteria for Protection Systems for Nuclear Power Generating Stations) and/or IEEE-308(Criteria for Class 1E Electrical systems for Nuclear Power Generating Stations)applies, single failures of both active and passive electrical components were considered in recognition of the higher anticipated failure rates of passive electrical components relative to passive mechanical components;

m. Provisions have been made for control of active components of nuclear safety systems and ESF from the control room;

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n. Nuclear safety systems and ESF are designed to permit demonstration of their functional performance requirements;

o. The design of nuclear safety systems and ESF includes allowances for natural environmental disturbances such as earthquakes, tornadoes, floods, and storms at the site;

p. Standby electrical power sources have sufficient capacity to power all nuclear safety systems and ESF requiring electrical power;

q. Standby electrical power sources are provided to allow prompt reactor shutdown and removal of transient heat under circumstances where offsite power sources are not available;

r. Features of the plant that are essential to the mitigation of accident consequences are designed, fabricated, and erected to qualify standards that reflect the importance of the safety action to be performed;

s. A primary containment has been provided that completely encloses the reactor system. The primary containment employs the pressure suppression concept;

t. The primary containment is designed to retain integrity as a radioactive material barrier during and following accidents that release radioactive material into the primary containment volume;

u. It is possible to test primary containment integrity and leaktightness at periodic intervals;

v. A secondary containment has been provided that completely encloses both the primary containment and fuel storage areas. The secondary containment includes the standby gas treatment (SGT) system for controlling release of radioactive materials leaking from the primary containment in the event of an accident and also has the capability for filtering radioactive materials directly from the primary containment atmosphere during shutdown conditions;

w. The secondary containment has been designed to act as a radioactive material barrier, if required, when the primary containment is open for expected operational purposes;

x. The primary containment and secondary containment, in conjunction with other ESF, limit radiological effects of accidents resulting in the release of radioactive material to the containment vessel to significantly less than 10 CFR 50.67 limits;

y. Provisions have been made for removing energy from within the containment vessel as necessary to maintain the integrity of the containment system following accidents that release energy to the primary containment;

z. Piping that penetrates the primary containment structure and could serve as a path for the uncontrolled release of radioactive material to the environs is automatically isolated whenever such uncontrolled radioactive material release is threatened. Such isolation shall be effected in time to limit radiological effects to less than specified acceptable limits;

aa. Emergency Core Cooling Systems (ECCS) are provided to limit reactor temperature to temperatures below the onset of fragmentation in the event of a loss-of-coolant-accident (LOCA);

bb. The ECCS provide for continuity of core cooling over the complete range of postulated break sizes in the RCPB and are redundant;

cc. Operation of the ECCS is initiated automatically when required, regardless of the availability of offsite power supplies and the normal generating systems of the plant;

dd. The control room has been shielded against radiation and provided with a high efficiency filtration system so that continued occupancy under accident conditions is possible;

ee. In the event that the control room becomes inaccessible, it is possible to bring the reactor from power range operation to cold shutdown conditions by utilizing the local controls and equipment that are available outside the control room on the remote shutdown control panels;

ff. Backup reactor shutdown capability has been provided independent of normal reactivity control provisions. This backup system has the capability to shut down the reactor from any normal operating condition and subsequently to maintain the shutdown condition; and

gg. Fuel handling and storage facilities are designed to prevent inadvertent criticality and to maintain adequate shielding and cooling of fuel.

1.2.1.2 System Criteria

The principal design criteria for particular systems are listed in the following subsections.

1.2.1.2.1 Nuclear System Criteria

a. The reactor is designed to retain integrity as a radioactive material barrier throughout the design power range. The reactor is designed to accommodate, without loss of integrity, the pressures generated by the fission gases released from the fuel material throughout the design life of the fuel;

b. The fuel cladding, in conjunction with other plant systems, is designed to retain integrity throughout any abnormal operational transient;

c. Those portions of the nuclear system that form part of the RCPB are designed to retain integrity as a radioactive material barrier following abnormal operational transients and accidents;

d. Heat removal systems are provided in sufficient capacity and operational adequacy to remove heat generated in the reactor for the full range of normal operational conditions from plant shutdown to design power and for any abnormal operational transient. The capacity of such systems is adequate to prevent reactor damage;

1.2.1.2.2 Power Conversion System Criteria

Components of the power conversion system have been designed to perform the following basic objectives.

a. Produce electrical power from the steam exiting from the reactor, condense the steam into water, and return water to the reactor as feedwater, with a major portion of its gaseous and particulate impurities removed; and

b. Ensure that any fusion products or radioactivity associated with the steam and condensate during normal operation are safely contained inside the system or are released under controlled conditions in accordance with waste disposal procedures.

1.2.1.2.3 Electrical Power Systems Criteria

Sufficient offsite and onsite standby sources of electrical power are provided to attain prompt shutdown and continued maintenance of the plant in a safe shutdown under all credible circumstances. The power sources are adequate to accomplish all required engineered safety feature functions under postulated design basis accident conditions.

1.2.1.2.4 Radwaste System Criteria

a. The gaseous and liquid radwaste systems are designed to limit the release of radioactive effluents from the plant during normal operation within those limits specified in 10 CFR 20 and 10 CFR 50, Appendix I;

b. The solid radwaste disposal system is designed so that during normal operation offsite shipments will be in accordance with applicable regulations, including 10 CFR 20, 10 CFR 71, and 49 CFR 171 through 10 CFR 179, as appropriate; and

c. The design of the systems provide means by which plant operations personnel are alerted whenever operational limits on the release of radioactive material are approached.

1.2.1.2.5 Auxiliary Systems Criteria

a. Fuel handling and storage facilities are designed to prevent criticality and to maintain adequate shielding and cooling for spent fuel. Provision is made for maintaining the cleanliness of spent fuel cooling and shielding water;

b. Other auxiliary systems, such as standby service water (SW), fire protection (FP) heating ventilation and air conditioning (HVAC), communications, and lighting systems are designed to function during normal, abnormal, and/or accident conditions; and

c. Auxiliary systems that are not required to effect safe shutdown of the reactor or maintain it in a safe shutdown condition are designed such that failures of these systems shall not prevent the essential auxiliary systems from performing their design functions.

1.2.1.2.6 Shielding and Access Control Criteria

a. Radiation shielding is provided and access control patterns are established to allow a properly trained operating staff to control radiation doses within the limits of published regulations in any normal mode of plant operation; and

b. The control room is shielded against radiation and has a high efficiency filtration system, so that occupancy is possible under accident conditions and TEDE doses are less than those set by Criterion 19 of 10 CFR Part 50, Appendix A and 10 CFR 50.67.

1.2.1.2.8 Nuclear Safety Systems and ESF Criteria

Principal design criteria for nuclear safety systems and ESF correspond to criteria j through q, aa through cc, and ee through ff in Section 1.2.1.1.2.

1.2.1.3 Plant Design Criteria

The plant design criteria are based on general design criteria given in Appendix A of 10 CFR Part 50. Conformance to these criteria is discussed in Section 3.1. The classification of structures, components, and systems is discussed in Section 3.2.

The principal regulations are codes that are used extensively in plant design are highlighted in Table 1.2-1. Note that the codes listed may not be applicable in their entirety. The many codes and regulations applicable to individual systems or structures are discussed throughout the FSAR.

The plant shielding and radiation zone classification can be found in Table 1.2-2. Chapter 12 provides further details.

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Table 1.2-1

Principal Regulations and Codes Followed in Plant Design

Number	Title
10 CFR series	Code of Federal Regulations, principally:
10 CFR 20	Standards for Protection Against Radiation
10 CFR 50	Licensing of Production and Utilization Facilities
10 CFR 50, Appendix A	General Design Criteria for Nuclear Power Plant Construction Permits
10 CFR 50, Appendix B	Quality Assurance Criteria
10 CFR 50, Appendix I	Numerical Guides for Design Objectives and Limiting Conditions for Operation to Meet the Criterion “As Low As Is Reasonably Achievable”
10 CFR 100	Reactor Site Criteria
IEEE-279	IEEE Criteria for Nuclear Power Generating Station Protection Systems
IEEE-308	IEEE Criteria for Class IE Electrical Systems for Nuclear Power Generating Stations
ASME B&PV	ASME Boiler and Pressure Vessel Code:
AEC Press Release IN-817	Tentative Regulatory Supplementary Criteria for ASME Code-Constructed Pressure Vessels

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Table 1.2-2

Plant Shielding and Zone Classification

Zone	Description	Design Dose Rate (mrem/hr)
A	Uncontrolled, unlimited access	≤1.0
B	Controlled, unlimited access	≤1.0
C	Controlled, limited access	≤1.0
D	Controlled, limited access	≤2.5
E	Controlled, occupancy for short periods	≤100
F	Controlled, occupancy for short periods	≤100
G	For very short periods. Secured and controlled entrance	> 100

NOTES:

1. Radiation Zone A areas can be occupied by plant personnel or visitors for unlimited periods.
2. Radiation Zone B areas are areas where whole body dose is not expected to exceed 1.25 rem per calendar quarter.
3. Areas having dose rates in excess of 100 mrem/hr are posted as high radiation areas and access is secured and controlled.
4. Radiation Zone D, E and F areas can be entered only after the radiation level is determined, the working time limit is established, and an RWP is issued.
5. Accessible areas have dose rates of less than 100 mrem/hr.
6. Access to all controlled areas is through controlled check points.
7. Controlled and limited access areas are identified by warning signs.